COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 / Automating Server Management	**SECTION:**	CPE31S2
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SKILLS EXAM	**PRELIMS**

1. Setup

The laboratory environment consists of:
- **1 Ubuntu Control Node**
- **2 Ubuntu Managed Nodes**
- **1 CentOS Managed Node**

Replace the sample IP addresses below with the actual IP addresses of your virtual machines.

Hostname	Operating System	IP Address
server1	Ubuntu	192.168.56.105
server2	Ubuntu	192.168.56.106
CentOS	CentOS	192.168.56.107

2. MATERIALS AND EQUIPMENT

- VirtualBox
- Personal Computer

3. PROCEDURE

Part A: Installing and Verifying Ansible

1. Update the package repository.
2. Install pip if it is not yet installed.
3. Install Ansible.
4. Verify that Ansible is installed correctly.
5. Verify SSH connectivity to each managed node.

Submit a screenshot of the following:
- Ansible version
- SSH connectivity

```
ace@workstation:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/ace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

```
ace@server1:~$ exit
logout
Connection to 192.168.56.105 closed.
```

```
ace@server2:~$
logout
Connection to 192.168.56.106 closed.
```

```
ace@workstation:~$ ssh ace@192.168.56.107
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Tue Aug 18 08:36:30 2026 from 192.168.56.104
[ace@centos ~]$
[ace@centos ~]$
logout
Connection to 192.168.56.107 closed.
ace@workstation:~$
```

Part B: Building an Inventory File

Inventories provide Ansible with information about managed nodes and their network locations.

1. Create a working directory.
2. Create an inventory file.
3. Add the following content.
4. Verify the inventory structure.
5. Ping all managed nodes.

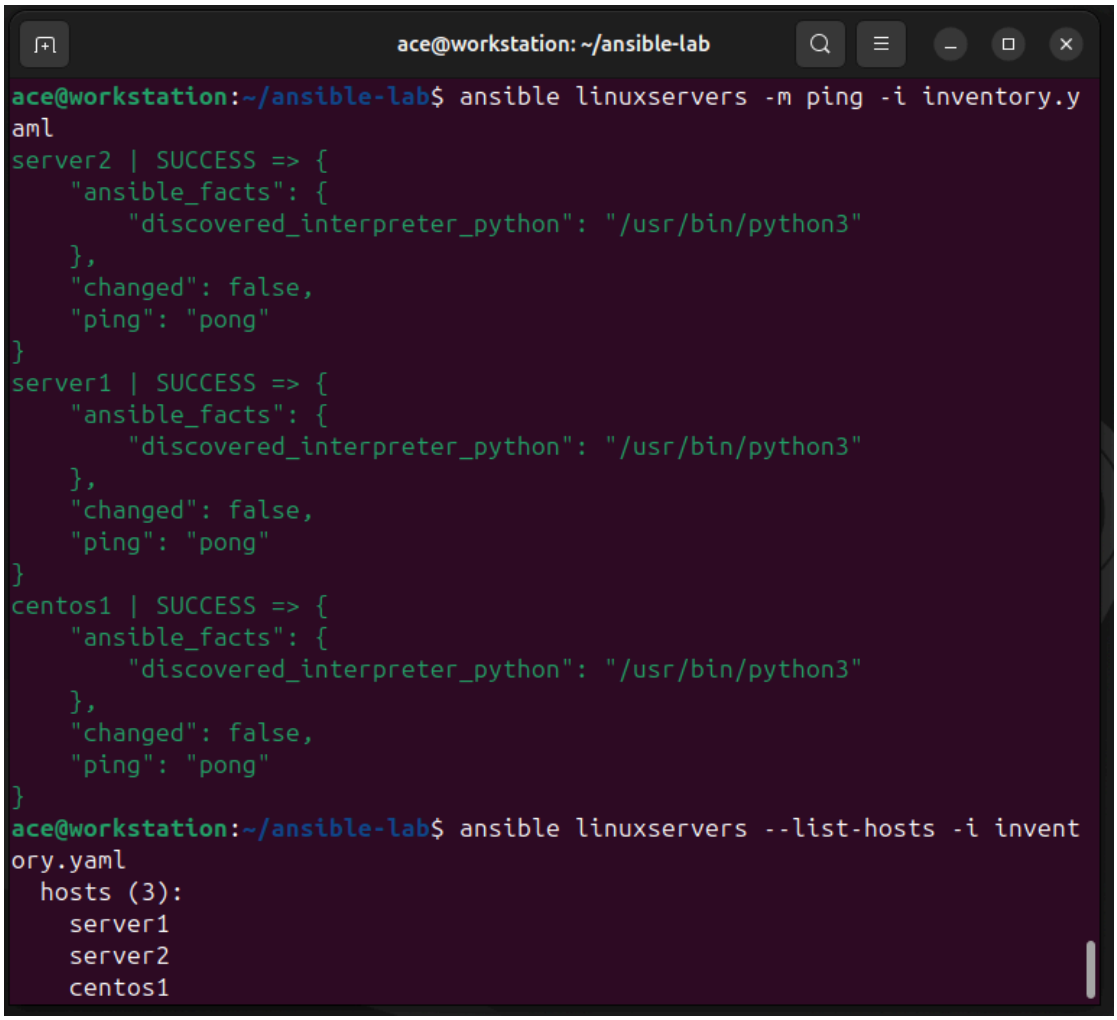
   Demonstrate the following:
   - inventory.yaml
   - Successful inventory verification
   - Successful ping results

```
ace@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Part C: Creating a Metagroup
Metagroups allow multiple groups to be combined into a single logical group.

1. Modify the inventory file.
2. Test the metagroup.
3. Display all hosts belonging to the metagroup.

   Demonstrate successful execution of the metagroup.



Part D: Creating Variables

Variables allow information such as operating systems and usernames to be assigned to hosts and groups.

1. Add host variables.
2. Add group variables.
3. Create a playbook to display variables.
4. Insert the following content.
5. Run the playbook.

   Demonstrate successful use of variables.

```
TASK [Gathering Facts] ************************************
**********************
ok: [server1]
ok: [server2]
ok: [centos1]

TASK [Show Operating System] ******************************
**********************
ok: [server1] => {
    "msg": "Host server1 is running Ubuntu in lab environme
nt"
}
ok: [server2] => {
    "msg": "Host server2 is running Ubuntu in lab environme
nt"
}
ok: [centos1] => {
    "msg": "Host centos1 is running CentOS in lab environme
nt"
}

PLAY RECAP ************************************************
**********************
centos1                    : ok=2    changed=0    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=2    changed=0    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0

ace@workstation:~/ansible-lab$
```

Part E: Creating Your First Playbook
Playbooks are YAML files used to automate tasks on managed nodes.

1. Create a playbook.
2. Add the following content.
3. Execute the playbook.
4. Observe the results. Successful tasks should display an ok status and a play recap summary.

Demonstrate:
- Successful playbook execution
- Successful task completion on all managed nodes

```
TASK [Display uptime] ***********************************
*********************
ok: [server1] => {
    "msg": " 02:06:00 up 46 min,  2 users,  load average: 0
.00, 0.03, 0.00"
}
ok: [server2] => {
    "msg": " 02:06:00 up 39 min,  2 users,  load average: 0
.01, 0.02, 0.00"
}
ok: [centos1] => {
    "msg": " 02:06:00 up 50 min,  2 users,  load average: 0
.00, 0.00, 0.00"
}

PLAY RECAP **********************************************
*********************
centos1                    : ok=3    changed=1    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=3    changed=1    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=3    changed=1    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0
```

4. SUPPLEMENTARY

Create a new group named servers that contains only:
- ubuntu1
- centos1

Create a playbook named challenge.yaml that displays the message

```
None
Welcome to Ansible Laboratory
```

Run the playbook and demonstrate that only hosts in the servers group receive the message.

```
PLAY [Challenge Playbook] ***********************************
**********************

TASK [Gathering Facts] ***************************************
**********************
ok: [server1]
ok: [centos1]

TASK [Print welcome message] *********************************
**********************
ok: [server1] => {
    "msg": "Welcome to Ansible Laboratory"
}
ok: [centos1] => {
    "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP ***************************************************
**********************
centos1                    : ok=2    changed=0    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=2    changed=0    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0

ace@workstation:~/ansible-lab$
```

Guide Questions

1. What is the role of the Control Node in Ansible?
   - This works through ansible handling playbooks, inventory, and configurations.
2. Why is an inventory file required?
   - It is to organize the hosts into groups in order to be easy to handle, and act as a place where ansible run tasks.
3. What is the purpose of a metagroup?

- Help to manage groups reducing the instance of having duplicates.

4. How do variables simplify Ansible administration?
   - Able to store reusable values, make some configurations simple and accessible across the system providing efficiency, and ease of control.
5. What is the difference between an ad hoc command and a playbook
   - Ad hoc only for fast and single tasks, while playbook is reusable and muchstructured.

5. CONCLUSIONS AND LEARNINGS

I learned that inventory files organize hosts into groups like ubuntu, centos, and custom groups like servers. Ansible lets me control many servers from one place using SSH. Inventory files group hosts like ubuntu, centos, and servers. Playbooks run the same tasks on multiple servers at once. This activity taught me that correct YAML formatting is important for Ansible to work properly.

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